

LIFE-CRAFT: A Multi-Agentic Conversational RAG Framework for Lifestyle Medicine Coaching with Context Traceability and Case-Based Evidence Synthesis

Hania Aslam ¹[0009-0001-2088-9068], Gousia K. Malak ¹, Max Renault¹ and Rajat M. Thomas¹

¹ Weill Cornell Medicine-Qatar, Doha, Qatar
haa4018@qatar-med.cornell.edu

Abstract. Lifestyle medicine offers a proactive and behavior-centric approach to managing chronic diseases, yet scalable, personalized coaching remains a significant challenge. We present LIFE-CRAFT (Lifestyle Intervention Facilitation Engine-Conversational RAG Framework with Traceability), a novel multi-agentic framework that leverages Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG) for personalized, lifestyle medicine related guidance. Central to LIFE-CRAFT is an orchestrator agent that interprets user queries and dynamically routes them to specialized supporting agents with expertise in prominent subdomains of lifestyle medicine. LIFE-CRAFT distinguishes itself from prior LLM-based assistants by introducing adaptive RAG pipeline with specialized sub-domain expert agents alongside dynamic domain routing and iterative feedback-based self-correcting refinement mechanisms for fine-grained context traceability as each agent’s response is grounded in semantically relevant document chunks retrieved from a well-curated, expert-reviewed PDF corpus of over 7,000 PubMed articles. For every generated response, the system logs the source document context that contributed to the response, providing transparency and explainability crucial for health-related decision support. The framework is evaluated across 864 GPT-4-generated lifestyle case scenarios using both LLM-based and human-in-the-loop evaluation. Initial results indicate that LIFE-CRAFT achieves higher context alignment and perceived trustworthiness, owing to its modular architecture, agentic decision-making, and transparent evidence grounding. This work demonstrates the feasibility and promise of multi-agentic, traceable RAG systems in supporting scalable, explainable, and personalized digital lifestyle medicine coaching—laying the foundation for future health AI systems that blend reasoning, retrieval, and real-world relevance. We intend to release the code in a GitHub repository upon acceptance of the paper to promote transparency and reproducibility.

Keywords: Lifestyle Medicine, Large Language Models, Explainable AI, Multi-Agent Systems, Digital Coaching, Traceability

1 Introduction

The rising global prevalence of lifestyle-related health conditions such as obesity, diabetes, cardiovascular disease, and mental health disorders has led to increased demand for accessible, accurate, and personalized health information. However, access to expert guidance in lifestyle medicine remains limited due to various socioeconomic and psychological barriers. Individuals often hesitate to seek in-person consultations due to stigma, cost, time constraints, or lack of immediate availability of trained professionals. In this context, the intersection of lifestyle medicine and intelligent, AI-powered tools may serve as a transformative opportunity to bridge a critical gap in care delivery and self-guided wellness.

We present a novel multi-agentic RAG-framework specifically tailored for the lifestyle medicine domain. While prior research has explored the use of RAG technique for knowledge-grounded response generation [1,2] and multi-agent frameworks [3,4] for complex reasoning and task decomposition in domains such as drug discovery[5,6], scientific research [7-9] and customer support [10], the application of these approaches to lifestyle medicine remains largely unexplored. This gap is especially significant given the rising burden of lifestyle-related conditions such as obesity, diabetes, and cardiovascular diseases. Our system addresses this need by acting as a virtual lifestyle coach, capable of delivering personalized, timely, and evidence-backed guidance grounded in behavioral science and lifestyle medicine literature. In contrast to general-purpose models such as GPT-4 or DeepSeek, which often rely on broad and sometimes outdated internet knowledge, our pipeline operates on a curated private repository of over 7,000 lifestyle medicine research PDFs, ensuring that responses are grounded in peer-reviewed, domain-specific evidence.

1.1 Key Innovations and Motivation:

- **Domain-Aware Agentic Architecture:** Directs queries to the most relevant lifestyle medicine subdomain expert agents.
- **Reduced Hallucination and Enhanced Credibility:** Mitigates hallucinations, a common issue in general-purpose LLMs.
- **Traceability and Evidence Linking:** Each generated answer includes source tracing, enhancing transparency and user trust.
- **Iterative Refinement:** The framework supports multi-turn refinement ensuring deeper coverage and tailored insights, unlike single-pass models.
- **Human and Automated Evaluation:** Dual-evaluation scheme with automated visual insights for LLM response verification and human-in-the-loop evaluations.

2 Related Work

Recent advances in retrieval-augmented generation (RAG) have significantly improved the performance of large language models on knowledge-intensive tasks. Lewis et al. [11] introduced the foundational RAG architecture by combining dense retrieval with generative modeling, enabling more factual and grounded responses in open-domain question answering. Subsequent works [12,13] have improved response quality through improved document fusion and advanced knowledge distillation techniques. Despite their effectiveness, these models typically operate in a monolithic fashion, relying on a single retrieval pipeline and lacking contextual adaptation or iterative refinement. Our system builds upon the RAG paradigm but introduces a multi-agent orchestration mechanism.

There has been growing interest in multi-agent LLM systems. Wu et al. [14] proposed AutoGPT, an early implementation of autonomous LLM agents capable of goal-driven task execution. AgentVerse [15] extended this paradigm by facilitating multi-agent collaboration, with agents specializing in different roles. Our system advances this line of work by embedding agentic reasoning within a transparent, domain-constrained RAG framework tailored for preventive healthcare.

In the healthcare domain, BioGPT [16], Med-PaLM [17], and ClinicalT5 [18] represent efforts to fine-tune or pretrain LLMs on biomedical corpora. However, these models are typically clinician-oriented and emphasize diagnostic accuracy over behavioral guidance. Within lifestyle medicine, digital health interventions have mostly centered around mobile health apps or behavior tracking tools [19,20]. To the best of our knowledge, no prior work has proposed a multi-agent, RAG-based architecture particularly within the lifestyle medicine domain, integrating features such as iterative refinement, contextual evidence visualization, and domain specific knowledge routing.

3 Methods

The LIFE-CRAFT framework is designed to address key limitations of monolithic LLM applications by introducing modularity, traceability, and expert domain routing. We expect it to pave the way for AI to act not just as an informant, but as a domain-guided, explainable advisor in preventive and lifestyle medicine within the growing digital health landscape.

3.1 LIFE-CRAFT Framework Architecture:

We propose and implement LIFE-CRAFT, a multi-agent Retrieval-Augmented Generation (RAG) framework tailored for lifestyle medicine applications. The overall architecture is presented in Fig. 1. Users interact with the system via natural language queries through the user interface. To manage these inputs effectively, our architecture begins with an orchestrator agent, referred to as the Lifestyle Medicine Expert Coach. This agent receives the initial query, performs structured decomposition using PydanticAI

for schema validation and consistency, and transforms the query into a format suitable for downstream retrieval.

The transformed query is forwarded to a dedicated router agent, which implements a hybrid routing algorithm combining lightweight keyword heuristics with embedding-based semantic relevance. Initially, a set of keyword-to-agent mappings (e.g., “diet” → Weight Management) provides rapid filtering. This is followed by a semantic layer where the query is embedded using the text-embedding-ada-002 model and compared to representative centroids of each agent’s domain corpus (vectorized using pgvector in Supabase). Based on this two-stage routing, the router non-deterministically selects the most relevant subset of domain significantly reducing computation overhead and aligns RAG performance with semantic specificity.

Before document retrieval, the system enters a Query Structuring Phase, where the natural language input is converted into structured semantic templates. This includes optional metadata tagging to improve filtering accuracy and reduce noise. Once agents return their outputs, a Corrective RAG Pipeline is invoked to re-rank retrieved chunks based on confidence scores and source quality. If the synthesized response is judged insufficient or ambiguous, the orchestrator may reactivate one or more agents with a reformulated. The resulting composite response is passed to an evaluation module, where outputs undergo LLM-based review for coherence, factual grounding, and source traceability. Final responses are delivered to the user as interactive/visual explanations, completing the loop. This engineered pipeline is orchestrated using LangGraph, that handles agent selection, conditional execution, retry logic, and final aggregation, enabling multi-agent orchestration.

To ensure practical deployment feasibility, LIFE-CRAFT is designed for scalable performance across key stages: routing, retrieval, and agent coordination. Retrieval is handled independently by each agent using vector search with pgvector’s HNSW indexing (approx. $O(\log N)$, where N is chunk count), enabling parallel, low-latency execution. LLM prompt lengths are constrained by selecting top-k chunks per agent, while recursive corrections only reinvoke underperforming agents.

In our current system, we partially mitigate hallucination and conflicting evidence by applying a relevance-based chunk ranking and aggregation mechanism during the Corrective RAG phase. Retrieved documents are first scored by semantic similarity, then filtered by metadata such as publication type and recency. If multiple agents return overlapping but non-identical insights, the orchestrator agent uses LLM-based summarization to consolidate responses, prioritizing consensus patterns while flagging divergences in reasoning chains. However, explicit detection and resolution of epistemic conflict is not yet automated.

3.2 Knowledge Base Curation Process

A key contribution of our work is the creation of an enriched lifestyle medicine knowledge base, demonstrating AI-human synergy as shown in Fig.2. Initially, domain experts conducted a human-in-the-loop screening of high-quality, peer-reviewed PubMed literature, focusing on evidence-based guidelines. The curated corpus underwent

thematic analysis and topic modeling, identifying six core domains aligned with agent specialization. In the second pass, GPT-4 generated synthetic case scenarios and QA pairs to simulate real-world queries, which were used as prompts in Perplexity AI for citation-driven expansion. The final corpus included 7,484 high-relevance documents, segmented into 200–400-word semantic chunks, tagged with domain metadata, embedded using OpenAI’s text-embedding-3-small model, and stored in Supabase tables.

We acknowledge potential biases inherent in PubMed-based sources. While our multi-agent approach supports thematic diversity, it doesn’t ensure epistemic balance. Future iterations will incorporate multilingual and regional sources, bias-aware sampling, and collaborative input from underrepresented public health practitioners.

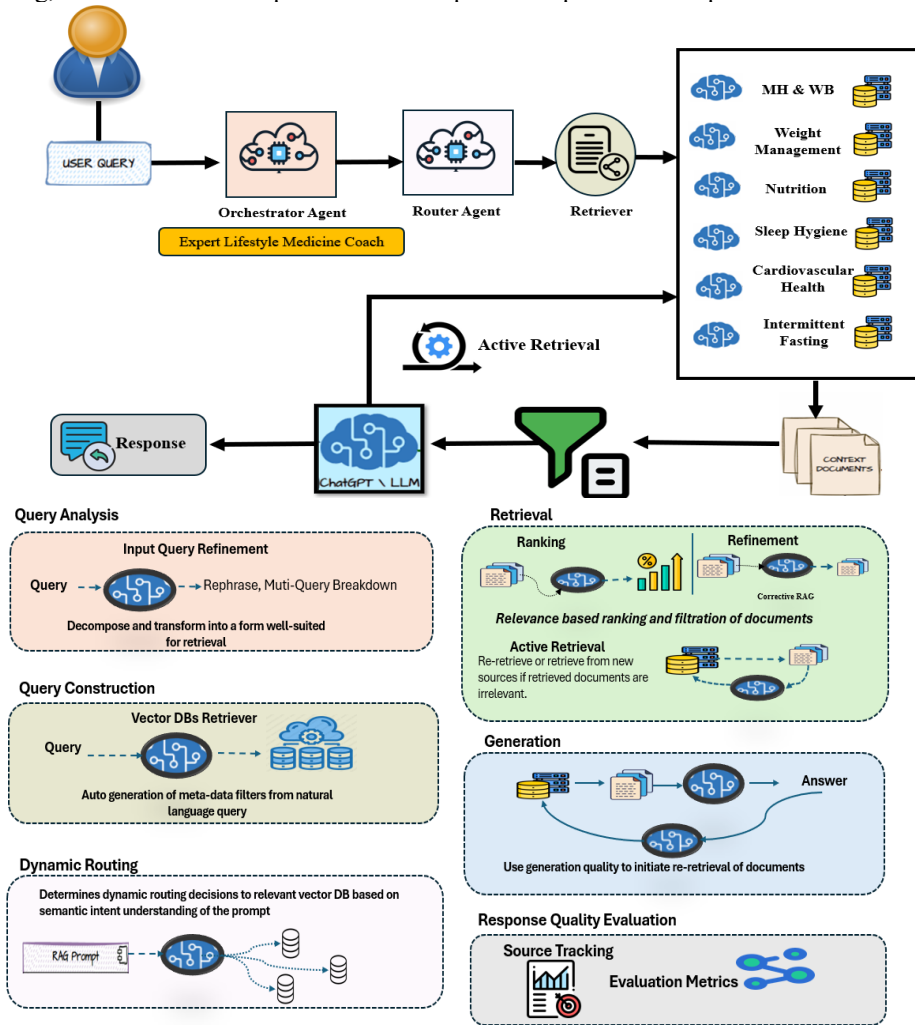


Fig. 1. The framework architecture for LIFE-CRAFT with color coded module summarization boxes.

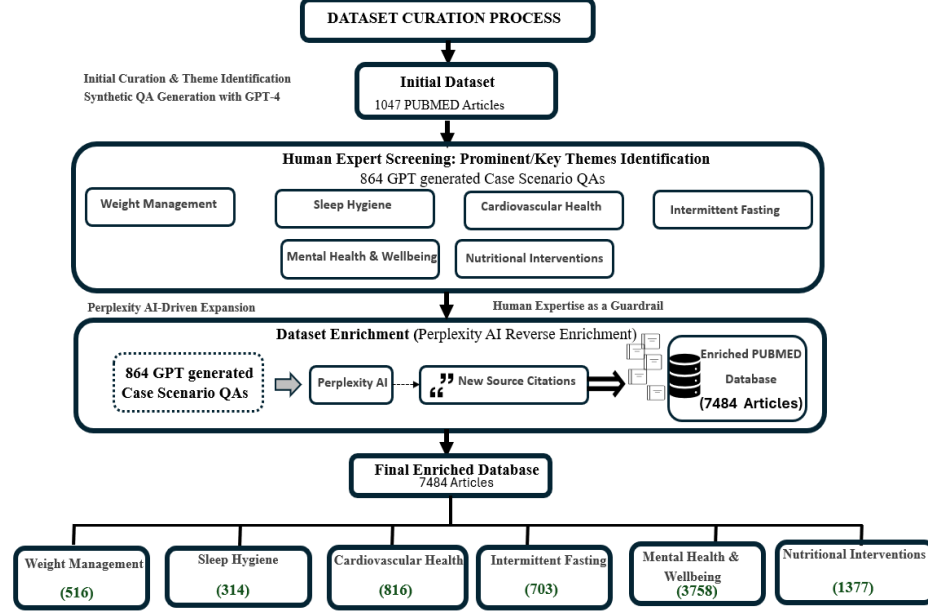


Fig. 2. Knowledge base curation strategy for RAG enabled response generation

4 Evaluation and Results:

We developed a simple user interface for query input, response display, and evaluation statistics. Fig. 3 shows our hybrid evaluation strategy: user prompts are processed through the agentic response pipeline, generating outputs shown to the user and evaluated by an LLM module for quality, transparency, and explainability.

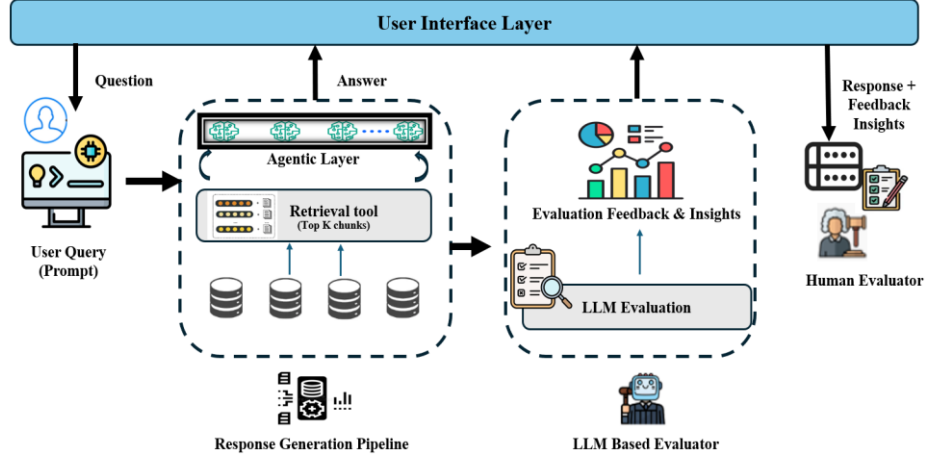


Fig. 3. Dual-evaluation strategy for quality assessment of the generated responses

Evaluated tasks include: (1) factual QA grounded in domain literature, (2) multi-agent query resolution requiring routing and aggregation, and (3) reasoning-intensive cases involving contextual interpretation and intervention prioritization.

Table 1. LLM based evaluation metrics for assessment of generated responses

Metric	Purpose	Formula	Interpretation
Completeness (C)	Measures coverage of key concepts in retrieved evidence	$C = \left(\frac{\sum_{i=1}^n \mathbb{1}(k_i \in \text{RetrievedChunks})}{n} \right) \times 100\%$	Higher values indicate better conceptual coverage
Confidence (Conf)	Quantifies semantic alignment between query and retrieved content	$\text{Conf} = \left(\frac{1}{m} \sum_{j=1}^m \text{sim}(q, c_j) \right) \times 100\%$	Reflects retrieval relevance and LLM trust level
Impact (I)	Reflects value of response based on importance or actionability	$I = \frac{1}{k} \sum_{i=1}^k w_i \cdot r_i \quad (\text{normalized})$	Assesses usefulness for health decision-making
Diversity (D)	Measures semantic spread among retrieved chunks	$D = \left(1 - \frac{1}{m(m-1)} \sum_{i=1}^m \sum_{j=i+1}^m \text{sim}(c_i, c_j) \right) \times 100\%$	Higher values indicate richer, non-redundant evidence

The metrics used for LLM based evaluation are shown in Table.1. For human-in-the-loop evaluation, two independent experts rated the three AI-generated responses (R1: GPT-4, R2: Perplexity AI, R3: LIFE-CRAFT) per question for a sampled subset of the case scenario QA database using a five-point rubric across five dimensional criteria.

We employed Cohen’s kappa coefficient to assess inter-rater agreement, and the averaged scores are presented in the accompanying plot (Fig. 4). These findings highlight the effectiveness of incorporating agentic RAG and explainability mechanisms in enhancing model performance. To strengthen robustness, we plan to expand the study to include a more diverse panel of evaluators, broaden the query set, and incorporate real-world user feedback in future iterations.

Although our evaluation includes baseline comparisons with state-of-the-art general-purpose LLM systems (GPT and Perplexity), we acknowledge the absence of direct benchmarking against existing digital health coaching platforms such as Omada, Noom, or Ada Health. This limitation stems partly from the proprietary nature and closed-source architecture of many such systems, which restrict standardized comparative analysis. However, future work will explore benchmarking against available academic or open-source digital health assistants.

5 Conclusion & Future Work

In this paper we propose a novel framework designed at the intersection of powerful AI tools and lifestyle medicine domain that may serve as a virtual coach. As a part of our

future work, we aim to integrate real-time data from wearable devices [21], to further enrich the contextual understanding of user behaviors and physiological states. This will allow our system to generate even more personalized and context-sensitive responses overcoming a common limitation in traditional consultations, where users may forget or misreport relevant lifestyle data. By seamlessly combining passive sensing with domain-aware reasoning, our platform has the potential to become an indispensable tool in preventive healthcare and self-managed wellbeing. Also, we intend to do more extensive evaluation while further enriching the data corpus across several lifestyle medicine sub domains for scaling and improvising our initial framework design. We are also working on a comprehensive human centric behavioral ontology framework to characterize personas and engineer the context of the LLMs for personalized responses.

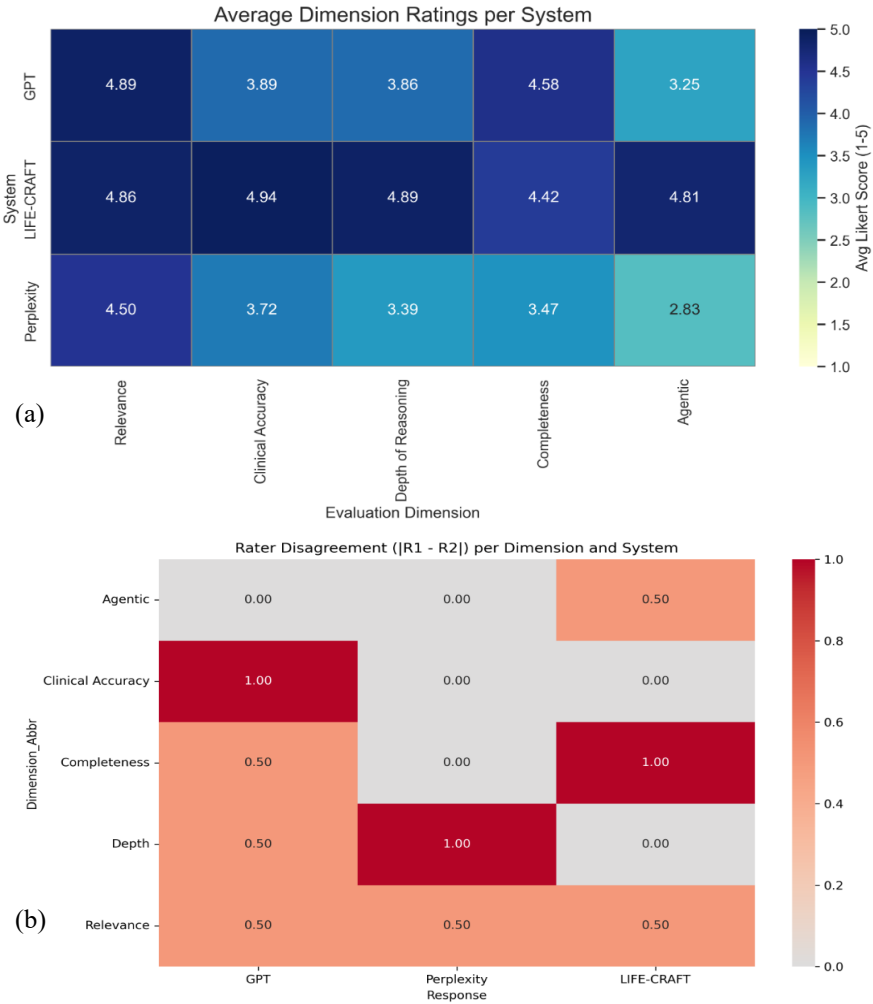


Fig. 4. (a) Human evaluation averaged scores summarized across five dimensions (b) Inter-rater reliability quantified via heatmap using Cohen's κ

We acknowledge that GPT-4 and Perplexity, while strong general-purpose baselines, are not specifically optimized for the lifestyle medicine domain. Their inclusion in our evaluation was motivated by their widespread adoption and competitive performance across general and medical question answering tasks, providing a realistic benchmark for assessing the added value of our domain-specialized, multi-agentic architecture. However, we agree that comparisons with domain-adapted models—such as Med-PaLM, ClinicalT5, or fine-tuned biomedical variants of LLaMA or Mistral—would offer a more rigorous assessment of relative performance. As part of future work, we intend to conduct targeted hallucination audits using established benchmarks (e.g., FaithDial, TruthfulQA) and manual annotation for factual consistency that may help to quantify hallucination reduction, we acknowledge that regulatory compliance considerations—such as those outlined by the FDA’s Software as a Medical Device (SaMD) guidelines and EU Medical Device Regulation (MDR)—are critical for any future real-world deployment. At present, the system is intended for informational, and decision-support use only and has not been validated for diagnostic or therapeutic purposes. These compliance pathways will be considered in collaboration with clinical stakeholders during the system’s translational roadmap.

Disclosure of Interests: The authors have no competing interests to declare that are relevant to the content of this article.

References

1. Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., Yih, W.T., Rocktäschel, T., Riedel, S., Kiela, D.: Retrieval-augmented generation for knowledge-intensive NLP tasks. In: Proceedings of the 34th International Conference on Neural Information Processing Systems (NeurIPS 2020), pp. 9459–9474. Curran Associates Inc., Red Hook (2020)
2. Izacard, G., Grave, E.: Leveraging passage retrieval with generative models for open-domain question answering. In: Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics (EACL), pp. 874–880. Association for Computational Linguistics, Online (2021)
3. Hong, S., Zheng, X., Chen, J., Cheng, Y., Wang, J., Zhang, C., Wang, Z., Yau, S.K.S., Lin, Z., Zhou, L., Ran, C.: MetaGPT: Meta programming for multi-agent collaborative framework. arXiv preprint arXiv:2308.00352 (2023)
4. Silva, M.A.L., de Souza, S.R., Souza, M.J.F., de Franca Filho, M.F.: Hybrid metaheuristics and multi-agent systems for solving optimization problems: A review of frameworks and a comparative analysis. Appl. Soft Comput. 71, 433–459 (2018)
5. Liu, S., Lu, Y., Chen, S., Hu, X., Zhao, J., Lu, Y., Zhao, Y.: DrugAgent: Automating AI-aided drug discovery programming through LLM multi-agent collaboration. arXiv preprint arXiv:2411.15692 (2024)
6. Ghafarollahi, A., Buehler, M.J.: ProtAgents: Protein discovery via large language model multi-agent collaborations combining physics and machine learning. Digit. Discov. 3(7), 1389–1409 (2024)
7. Barbosa, R., Santos, R., Novais, P.: Collaborative problem-solving with LLM: A multi-agent system approach to solve complex tasks using AutoGen. In: International Conference

on Practical Applications of Agents and Multi-Agent Systems, pp. 203–214. Springer, Cham (2024)

8. Ferrag, M.A., Tihanyi, N., Debbah, M.: From LLM reasoning to autonomous AI agents: A comprehensive review. *arXiv preprint arXiv:2504.19678* (2025)
9. Guo, T., Chen, X., Wang, Y., Chang, R., Pei, S., Chawla, N.V., Wiest, O., Zhang, X.: Large language model based multi-agents: A survey of progress and challenges. *arXiv preprint arXiv:2402.01680* (2024)
10. Shareef, F.: Enhancing conversational AI with LLMs for customer support automation. In: 2nd Int. Conf. on Self Sustainable Artificial Intelligence Systems (ICSSAS), pp. 239–244. IEEE (2024)
11. Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., Yih, W.T., Rocktäschel, T., Riedel, S.: Retrieval-augmented generation for knowledge-intensive NLP tasks. *Adv. Neural Inf. Process. Syst.* 33, 9459–9474 (2020)
12. Izacard, G., Grave, E.: Leveraging passage retrieval with generative models for open-domain question answering. *arXiv preprint arXiv:2007.01282* (2020)
13. Zhou, Y., Shen, T., Geng, X., Tao, C., Shen, J., Long, G., Xu, C., Jiang, D.: Fine-grained distillation for long document retrieval. In: *Proc. AAAI Conf. Artif. Intell.* 38(17), 19732–19740 (2024)
14. Yang, H., Yue, S., He, Y.: Auto-GPT for online decision making: Benchmarks and additional opinions. *arXiv preprint arXiv:2306.02224* (2023)
15. Chen, W., Su, Y., Zuo, J., Yang, C., Yuan, C., Qian, C., Chan, C.M., Qin, Y., Lu, Y., Xie, R., Liu, Z.: AgentVerse: Facilitating multi-agent collaboration and exploring emergent behaviors in agents. *arXiv preprint arXiv:2308.10848* (2023)
16. Luo, R., Sun, L., Xia, Y., Qin, T., Zhang, S., Poon, H., Liu, T.Y.: BioGPT: Generative pre-trained transformer for biomedical text generation and mining. *Brief. Bioinform.* 23(6), bbac409 (2022)
17. Lehman, E., Johnson, A.: Clinical-T5: Large language models built using MIMIC clinical text. *PhysioNet* (2023)
18. Alzeer, J.: Integrating medicine with lifestyle for personalized and holistic healthcare. *J. Public Health Emerg.* 7 (2023)
19. Calvaresi, D., Marinoni, M., Dragoni, A.F., Hilfiker, R., Schumacher, M.: Real-time multi-agent systems for telerehabilitation scenarios. *Artif. Intell. Med.* 96, 217–231 (2019)
20. Kim, Y., Xu, X., McDuff, D., Breazeal, C., Park, H.W.: Health-LLM: Large language models for health prediction via wearable sensor data. *arXiv preprint arXiv:2407.00000* (2024)
21. Ferrara, E.: Large language models for wearable sensor-based human activity recognition, health monitoring, and behavioral modeling: A survey of early trends, datasets, and challenges. *Sensors* 24, 5045 (2024). <https://doi.org/10.3390/s24155045>